■ CarbonSense

Carbon Footprint Tracker — Hack2Skill Challenge 3

Climate Tech & Sustainability | Submitted: June 2026

Problem Statement

Individuals today have little awareness of how their daily habits contribute to climate change. There is a need for a simple, personalized tool that helps people **understand, track, and reduce** their carbon footprint through simple actions and personalized insights.

Our Solution

CarbonSense is a web-based carbon footprint tracker that allows users to input their daily lifestyle habits and instantly see their CO_2 emissions — with personalized tips and insights to reduce them.

Key Features

Feature	Description
■ Transport Tracker	Car, bike, bus, and flight emission inputs
■ Home Energy Tracker	Electricity, LPG, and AC usage calculator
■■ Food & Diet Tracker	Meat, dairy, and food waste tracking
■■ Shopping Tracker	Clothes, electronics, and plastic bag impact
■ Live CO2 Gauge	Real-time annual footprint score with visual meter
■ Personalized Insights	Compared against India's average (1.9t/year)
■ Reduction Tips	Actionable suggestions per category
■ Green Pledge	Motivates users to commit to reducing 10%

Tech Stack

Layer	Technology
Frontend	React.js
Logic	JavaScript (Emission factor calculations)
Styling	CSS with responsive mobile-first design
Deployment	Vercel / Netlify (free hosting)

Impact & Benefits

✓ Helps users identify their biggest emission source instantly

✓ Compares personal footprint to India's national average (1.9t CO2/year)

✓ Educates users on the 1.5 degree C climate target (under 2t/person/year)
✓ Encourages behavior change through simple, actionable reduction tips
✓ Green Pledge feature drives long-term user commitment

Methodology

Emission factors are based on standard carbon accounting references including India's electricity grid emission factor (0.82 kg CO2/kWh), IPCC transport emission averages, and food lifecycle assessment data. All calculations are done client-side in real-time for instant feedback.